Literature Review: Art Gallery Problems for Orthogonal Polygons and Orthogonal Polyhedra

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Abstract

This review organizes the art-gallery literature most relevant to perimeter-type guard bounds for orthogonal objects. In two dimensions, the perimeter-over-six question is meaningful only after fixing a lattice scale: it is proven for hole-free polyominoes, conditional/open for polyominoes with holes, and open for general integral orthogonal polygons. In three dimensions, the standard term is *orthogonal polyhedron* or *orthogonal polytope*, not "orthonormal." A direct point-guard analogue of the planar theorem fails: orthogonal polyhedra may require $\Theta(n^{3/2})$ point guards. The natural 3D analogue therefore shifts to edge guards, reflex-edge guards, face guards, or discrete polycube visibility models, where several Urrutia-type conjectures remain open.

1 Reading Guide and Terminology

2D objects. An *orthogonal polygon* is a polygon whose edges are horizontal or vertical. A *polyomino* is a finite edge-connected union of unit axis-parallel squares; its perimeter is naturally the number of unit boundary edges, including hole boundaries when holes are present. An *integral orthogonal polygon* has integer coordinates, so its perimeter is a lattice length. An *ortho-unit polygon* is an orthogonal polygon all of whose edges have length one; in that model, the number of vertices equals the perimeter.

3D objects. The standard 3D term is *orthogonal polyhedron*: a polyhedron whose faces are axis-aligned, equivalently whose face normals are coordinate directions. In higher dimensions, the corresponding term is *orthogonal polytope*. The word *orthonormal* describes vectors or bases, not galleries. Polycubes are the voxel analogue of polyominoes and form a discrete subclass of orthogonal polyhedra.

Guard models. Unless a section says otherwise, a 2D guard is a standard point guard with straight-line visibility inside the polygon. In 3D the model must be stated: the literature distinguishes point guards, vertex guards, edge guards, reflex-edge guards, face guards, open edge guards, and discrete polycube visibility graph models. These models are not interchangeable.

Why perimeter needs a lattice scale. No theorem of the form "at most $P/6$ guards" can hold for arbitrary real-coordinate orthogonal polygons if P is Euclidean perimeter: scaling a polygon down preserves the guard number while making P arbitrarily small. The perimeter-over-six question is therefore a lattice-normalized question: polyomino perimeter, integral perimeter, or a fixed unit-edge model.

2 Flow of the Literature

1. Classical art-gallery theorems start with arbitrary simple polygons: Chvatal proves $\lfloor n/3 \rfloor$ and Fisk gives the standard triangulation proof [7, 11]. Shermer, O'Rourke, and Urrutia provide the main survey/book context [24, 20, 25].

2. Orthogonal polygons admit stronger vertex-count theorems: Kahn–Klawe–Kleitman prove the tight $\lfloor n/4 \rfloor$ bound for simple orthogonal polygons [15].

3. Holes complicate the 2D theory. Vertex-guard conjectures for orthogonal polygons with holes, especially Shermer's and Hoffmann's formulations, remain separate from the perimeter-over-six point-guard question [26, 14, 31].

4. Polyominoes introduce area and lattice-perimeter parameters. The tight area theorem allows holes [3], while Maßberg's perimeter-over-six theorem is proven for hole-free polyominoes and becomes conditional in the presence of holes [17, 18].

5. Recent ortho-unit work proves a strong unit-edge theorem and states the broader integral-perimeter conjecture [8].

6. In 3D, point guards are too weak for a planar-style theorem: orthogonal polyhedra need $\Theta(n^{3/2})$ point guards in the worst case [21, 27]. Edge-guard and reflex-edge-guard conjectures become the main 3D analogue [2, 29, 6].

3 Status by Model

Simple orthogonal polygons, point guards.
Proven tight: every n-vertex simple orthogonal polygon is guardable by $\lfloor n/4 \rfloor$ point guards [15, 13].

Orthogonal polygons with holes, vertex guards.
Several upper bounds and partial sharp results are known. O'Rourke gives the classical $\lfloor (n + 2h)/4 \rfloor$ style bound for h holes [20]. Shermer's stronger $\lfloor (n + h)/4 \rfloor$ conjecture is open in general; Zylinski gives a coloring proof of Aggarwal's theorem for $h \leq 2$ and cactus-dual quadrilateralizations [26, 31].

Hole-free polyominoes, point guards, lattice perimeter ℓ.
Proven tight: for $\ell \geq 6$, Maßberg proves that $\lfloor \ell/6 \rfloor$ point guards suffice [17]. Use $\max\{1, \lfloor \ell/6 \rfloor\}$ if the one-square case is included.

Polyominoes with holes, point guards, lattice perimeter ℓ.
Open/conditional. Maßberg's text explicitly states that the extension to holes would follow from a maximal-rectangle packing conjecture [18]. No later proof was found in this literature search, current to May 22, 2026.

Polyominoes with holes, point guards, area m.
Proven tight: every m-cell polyomino, holes allowed, is guardable by $\lfloor (m + 1)/3 \rfloor$ point guards [3]. This does not settle perimeter-over-six.

Ortho-unit polygons, point guards.
Proven tight: every ortho-unit polygon with $n \geq 12$ vertices is guardable by $\lfloor (n - 4)/8 \rfloor$ guards

[8]. Since $n = P$ in the ortho-unit model, this is stronger than $P/6$ for that narrow class.

Integral orthogonal polygons, point guards, perimeter N.
Open. Diaz-Banez et al. state the natural conjecture that $\lfloor N/6 \rfloor$ guards suffice [8]. No later proof was found in this literature search, current to May 22, 2026.

Polycubes and polyhypercubes.
Pinciu studies discrete polyform variants, including polycubes and polyhypercubes [22]. These are relevant for 3D lattice analogues, but they use a discrete polyform/visibility-graph framework rather than the continuous point-guard model for orthogonal polyhedra.

Orthogonal polyhedra, point guards.
No planar-style linear theorem exists in terms of vertices. The Paterson–Yao bound gives $\Theta(n^{3/2})$ point guards as sufficient and sometimes necessary for n-vertex orthogonal polyhedra [21, 27].

Orthogonal polyhedra, edge and reflex-edge guards.
The central open direction is an Urrutia-type edge-guard theorem: lower-bound examples require roughly $m/12$ edge guards, while known upper bounds are larger. Benbernou et al. improve the orthogonal upper bound to asymptotic $(11/72)m$ edge guards, or $(7/12)r$ in terms of reflex edges [2]. Viglietta proves optimal-style bounds for 2-reflex orthogonal polyhedra [29]. Cano, Toth, Urrutia, and Viglietta improve the general 3D polyhedron edge-guard bound to $(5/6)m$, while noting a substantial remaining gap [6].

4 Two-Dimensional Orthogonal Polygons

4.1 Classical Background

The original art-gallery theorem is due to Chvatal: $\lfloor n/3 \rfloor$ guards always suffice for a simple n-vertex polygon, and this is tight [7]. Fisk's short proof via triangulation and three-coloring became the standard proof template [11]. O'Rourke's book remains the standard comprehensive reference for the classical theory [20]; Shermer and Urrutia survey art-gallery and illumination problems, including variants and open problems [24, 25]. On the algorithmic side, Abrahamsen, Adamaszek, and Miltzow prove that the general art-gallery problem is $\exists\mathbb{R}$-complete, underscoring why restricted orthogonal and lattice models are usually treated separately [1].

4.2 Simple Orthogonal Polygons

Kahn, Klawe, and Kleitman prove the tight orthogonal theorem: $\lfloor n/4 \rfloor$ point guards suffice for every simple orthogonal polygon with n vertices [15]. Gyori gives a short proof [13]. This theorem is a vertex-count theorem, not a perimeter theorem. In a lattice setting the vertex count and the perimeter may be very different, which is why the polyomino and integral-perimeter questions are separate.

The proof tradition connects to decomposition, coloring, and perfect-graph methods. Motwani, Raghunathan, and Saran study covering orthogonal polygons with star polygons [19]; Worman and Keil develop decomposition and visibility-graph machinery for orthogonal art-gallery problems [30]. Bjørling-Sachs proves the tight $\lfloor n/4 \rfloor$ edge-guard bound for rectilinear polygons, a useful 2D reference point for later edge-guard models [5].

4.3 Holes

Holes are one of the main fault lines in the literature. For orthogonal polygons with h holes, classical vertex-guard bounds exist, but the strongest natural conjectures are not settled in general. Urrutia's open-problem list records Shermer's conjecture that an n-vertex orthogonal polygon with h holes should be guardable by $\lfloor (n+h)/4 \rfloor$ vertex guards, and Hoffmann's conjecture that $\lfloor 2n/7 \rfloor$ vertex guards should suffice for orthogonal polygons with holes [26]. Zylinski proves the $\lfloor (n+h)/4 \rfloor$ bound under a cactus-dual quadrilateralization condition and recovers Aggarwal's theorem for $h \leq 2$ [31]. Hoffmann and Kriegel prove graph-coloring consequences for rectilinear polygons with holes, including upper bounds for rectilinear prison-yard variants [14].

These hole results are relevant context, but they do not prove the perimeter-over-six statement. They usually concern vertex guards and vertex counts; the perimeter-over-six problem concerns point guards and lattice perimeter.

5 Polyomino and Perimeter Results

5.1 Hole-Free Polyominoes

Maßberg proves the direct perimeter theorem for hole-free polyominoes [17]. In the notation of that paper, if a hole-free polyomino has perimeter $\ell \geq 6$, then it can be guarded by at most $\lfloor \ell/6 \rfloor$ point guards. The proof uses maximal axis-parallel rectangles in a rectilinear gallery, relates a rectangle packing number to the guard number, and then bounds the packing by the perimeter. Comb-like examples show tightness.

5.2 Polyominoes with Holes

The area theorem does allow holes: Biedl, Irfan, Iwerks, Kim, and Mitchell prove that every m-cell polyomino can be guarded by $\lfloor (m+1)/3 \rfloor$ point guards, and the bound is tight [3]. This is the strongest clean theorem for arbitrary polyominoes with holes in the standard point-guard model, but it is an area bound rather than a perimeter bound.

For perimeter, the hole case remains conditional. Maßberg observes that the perimeter theorem would extend to holes if a maximal-rectangle packing conjecture were true [18]. The important point for a new paper is that this is *not* a proof of perimeter-over-six for holes. It identifies the missing structural statement.

5.3 Ortho-Unit and Integral Orthogonal Polygons

Diaz-Banez et al. prove that every ortho-unit polygon with $n \geq 12$ vertices can be guarded by $\lfloor (n-4)/8 \rfloor$ guards, and that the bound is tight [8]. Since every edge has unit length in an ortho-unit polygon, n is the perimeter. This solves a narrow unit-edge class more strongly than $P/6$.

The same paper frames the broader integral-perimeter conjecture: every integral orthogonal polygon of perimeter N should be guardable by $\lfloor N/6 \rfloor$ point guards. This remains the closest published statement of the general perimeter-over-six problem beyond hole-free polyominoes.

6 Three-Dimensional Orthogonal Polyhedra

6.1 Point Guards

The direct 3D point-guard analogue of planar art-gallery theorems fails. In orthogonal polyhedra with n vertices, Paterson and Yao's binary-space partition result implies a tight $\Theta(n^{3/2})$ point-guard bound [21]. Viglietta's thesis presents this as Theorem 3.7 and draws the key modeling conclusion: point guards do not yield a linear orthogonal-polyhedron art-gallery theorem [27]. Therefore a 3D "perimeter-over-six" theorem is not obtained by simply replacing polygons with orthogonal polyhedra and keeping point guards.

6.2 Edge and Reflex-Edge Guards

The 3D literature often treats edge guards as the analogue of planar vertex guards. Urrutia conjectured that an orthogonal polyhedron with m edges should be guardable by $m/12+O(1)$ closed edge guards, matching known lower bound examples up to an additive constant [27]. A related reflex-edge conjecture asks whether roughly $(r-g)/2+1$ reflex edge guards suffice, where r is the number of reflex edges and g is the genus.

Benbernou, Demaine, Demaine, Kurdia, O'Rourke, Toussaint, Urrutia, and Viglietta improve the known asymptotic upper bounds for orthogonal polyhedra: $(11/72)m$ edge guards suffice in terms of total edges, and $(7/12)r$ suffice in terms of reflex edges [2]. Viglietta later proves optimal-style bounds for 2-reflex orthogonal polyhedra, namely orthogonal polyhedra whose reflex edges occur in only two directions [29]. These are major partial results, but they do not settle the general 3-reflex orthogonal-polyhedron conjectures.

Cano, Toth, Urrutia, and Viglietta give a broad theorem for arbitrary polyhedra in three-space: every polyhedron with m edges can be guarded by at most $(5/6)m$ edge guards [6]. This improves the general-polyhedron edge-guard bound, but it still leaves a large gap from the conjectured $m/6$ style lower/upper behavior for general polyhedra and the $m/12+O(1)$ orthogonal-polyhedron conjecture.

6.3 Face Guards and Polycube Variants

Face guards are another 3D model. Viglietta studies face-guarding for several classes of polyhedra, including orthogonal polyhedra, 4-oriented polyhedra, and 2-reflex orthostacks [28]. Face guards are useful for mapping the 3D model space, but they are not a direct continuation of the point-guard perimeter-over-six question.

For lattice 3D objects, Pinciu's work on polycubes and polyhypercubes gives a discrete polyform line of results [22]. This is the natural place to look if the intended 3D project is voxel/discrete rather than continuous orthogonal-polyhedron visibility.

7 Quoted Open-Problem Evidence

The following short quotations are included only to document open status. Each quote is followed by the conclusion that should be drawn from it.

> "If this conjecture is true"
>
> — Maßberg, immediately before the claimed hole extension [18]

Status confirmed. Maßberg's hole extension is conditional on Conjecture 6.10, not proven. The conjecture concerns whether a packing of maximal rectangles bounds the guard number in rectilinear galleries with holes. No later proof was found in this literature search, current to May 22, 2026.

> "also true for polyominoes with holes"
>
> — Maßberg, conditional conclusion after Conjecture 6.10 [18]

Status confirmed. This quote must not be used without the preceding condition. The published theorem is hole-free; the hole case is a stated conditional extension.

> "guarded with at most $\lfloor N/6 \rfloor$ guards"
>
> — Diaz-Banez et al., integral-perimeter conjecture [8]

Status confirmed. This is a conjecture in the 2025 ortho-unit paper. The paper proves the ortho-unit theorem, not the full integral orthogonal polygon conjecture. No later proof was found in this literature search, current to May 22, 2026.

> "Floor[(n+h)/4] vertex guards"
>
> — Urrutia's page recording Shermer's orthogonal-holes conjecture [26]

Status confirmed. This is a vertex-guard conjecture for orthogonal polygons with holes. It is relevant to holes, but it is not the same as the point-guard perimeter-over-six question.

> "no linear upper bound"
>
> — Viglietta, discussing point guards in orthogonal polyhedra [27]

Status confirmed. This is the main reason 3D point guards should not be treated as a direct analogue of the 2D perimeter theorem.

> "Further reducing this gap is left as an open problem."
>
> — Cano, Toth, Urrutia, and Viglietta, on edge guards in 3-space [6]

Status confirmed. As of the 2022 general edge-guard paper, the gap between lower bounds and upper bounds for edge guarding polyhedra remains an explicit open problem.

> "Is there a constant-factor approximation for polyominoes with holes?"
>
> — Filtser et al., for k-hop visibility in polyominoes [10]

Status confirmed. This is a current open problem in a related discrete visibility model, not in standard continuous point-guard visibility.

8 Relevant Open Problems

1. **Perimeter-over-six for polyominoes with holes.** Let P be a polyomino with holes and total lattice perimeter ℓ, including all hole boundaries. Is P always guardable by $\max\{1, \lfloor \ell/6 \rfloor\}$ point guards?

   Open-status check: confirmed by Maßberg's conditional wording in the habilitation text; No later proof was found in this literature search, current to May 22, 2026.

2. **Maßberg's maximal-rectangle packing conjecture for holes.** In a rectilinear gallery with holes, does the maximum packing size of maximal rectangles upper-bound the number of point guards?

   Open-status check: source-confirmed as Conjecture 6.10 in Maßberg's text. A proof would extend the hole-free perimeter theorem to polyominoes with holes.

3. **Integral orthogonal polygon perimeter conjecture.** Is every integral orthogonal polygon of perimeter N guardable by $\lfloor N/6 \rfloor$ point guards?

   Open-status check: source-confirmed as the natural conjecture in the 2025 ortho-unit paper; No later proof was found in this literature search, current to May 22, 2026.

4. **Shermer's vertex-guard conjecture for orthogonal polygons with holes.** Is every n-vertex orthogonal polygon with h holes guardable by $\lfloor (n+h)/4 \rfloor$ vertex guards?

   Open-status check: Urrutia's open-problem list records it as open; Zylinski proves important partial cases, including Aggarwal's $h \leq 2$ theorem.

5. **Hoffmann's vertex-guard conjecture for orthogonal polygons with holes.** Does $\lfloor 2n/7 \rfloor$ always suffice for vertex guarding any orthogonal polygon with holes?

   Open-status check: Urrutia's page records this as a distinct open conjecture. It is vertex-count based and does not imply perimeter-over-six.

6. **If perimeter-over-six with holes is false, find the correction term.** Possible parameters include total perimeter, outer perimeter, number of holes, hole perimeter, reflex vertices, or rectangle-packing obstructions.

   Open-status check: this is a derived research direction, not a named source conjecture. It follows from the current gap in the hole case.

7. **3D Urrutia edge-guard conjecture for orthogonal polyhedra.** Is every genus-zero orthogonal polyhedron with m edges guardable by $m/12 + O(1)$ edge guards?

   Open-status check: recorded in Viglietta's thesis and subsequent edge-guard literature; partial upper bounds are known, but the general conjecture is not settled.

8. **3D reflex-edge conjecture for orthogonal polyhedra.** Is every orthogonal polyhedron with $r > 0$ reflex edges and genus g guardable by $\lfloor (r-g)/2 \rfloor + 1$ reflex-edge guards?

   Open-status check: Viglietta proves this for 2-reflex orthogonal polyhedra and states the general version as a conjecture.

9. **Close the general 3D edge-guard gap.** For arbitrary polyhedra with m edges, improve the current gap between lower-bound examples and the $(5/6)m$ upper bound.

   Open-status check: Cano, Toth, Urrutia, and Viglietta explicitly state that reducing this gap remains open.

10. **Choose the correct 3D analogue of perimeter.** For orthogonal polyhedra or polycubes, decide whether the right parameter is total edges, reflex edges, surface area, voxel count, genus, or a combination.

    Open-status check: not a single named theorem, but forced by the literature: point guards have a superlinear vertex bound, while edge/reflex edge and polycube models have different behavior.

11. **Algorithmic complexity and approximation for restricted orthogonal models.** Minimum guarding is NP-hard in many orthogonal settings [23, 16, 3, 9, 4]. Identify exact polynomial subclasses and approximation guarantees for polyominoes with holes, integral orthogonal polygons, polycubes, and orthogonal polyhedra.

    Open-status check: several hardness results are known, while recent work on k-hop polyomino visibility still asks for a constant-factor approximation for polyominoes with holes [10].

9 Journals and Venues to Monitor

Journals

- *Discrete & Computational Geometry*
- *Computational Geometry: Theory and Applications*
- *Journal of Computational Geometry*
- *International Journal of Computational Geometry & Applications*
- *Graphs and Combinatorics*
- *Algorithmica*
- *Journal of the ACM*
- *ACM Transactions on Algorithms*
- *SIAM Journal on Discrete Mathematics*
- *SIAM Journal on Computing*
- *Journal of Computer and System Sciences*
- *Journal of Algorithms* and successor venues
- *The Electronic Journal of Combinatorics*
- *Discrete Applied Mathematics*
- *Information Processing Letters*
- *Journal of Discrete Algorithms*
- *Theoretical Computer Science*
- *Mathematical Logic Quarterly*
- *Electronic Notes in Discrete Mathematics*

Conferences and Workshops

- Symposium on Computational Geometry (SoCG)
- European Workshop on Computational Geometry (EuroCG)
- Canadian Conference on Computational Geometry (CCCG)

- Latin American Symposium on Theoretical Informatics (LATIN)
- International Symposium on Algorithms and Computation (ISAAC)
- European Symposium on Algorithms (ESA)
- ACM-SIAM Symposium on Discrete Algorithms (SODA)
- Workshop on Algorithms and Data Structures (WADS)
- Workshop on Graph-Theoretic Concepts in Computer Science (WG)
- EuroComb and related discrete-mathematics meetings

10 Annotated Source List

Foundational Art-Gallery Theory

Chvatal [7] and Fisk [11].
Foundational $\lfloor n/3 \rfloor$ theorem for simple polygons and the short triangulation/coloring proof.

O'Rourke [20].
Standard book reference for art-gallery theorems, algorithms, visibility, decompositions, and early 3D context.

Shermer [24] and Urrutia [25].
Survey references for art-gallery and illumination problems, useful for terminology, variants, and classical open-problem context.

Abrahamsen–Adamaszek–Miltzow [1].
Modern complexity reference proving $\exists\mathbb{R}$-completeness for the general art-gallery problem.

2D Orthogonal Polygons and Holes

Kahn–Klawe–Kleitman [15].
Primary source for the tight $\lfloor n/4 \rfloor$ theorem for simple orthogonal polygons.

Gyori [13].
Short proof of the rectilinear art-gallery theorem.

Hoffmann–Kriegel [14].
Graph-coloring result with consequences for rectilinear polygons with holes and prison-yard variants.

Zylinski [31].
Coloring proof of Aggarwal's theorem for orthogonal galleries with holes, including the $\lfloor (n+h)/4 \rfloor$ bound for $h \leq 2$ and cactus-dual cases.

Urrutia open-problem page [26].
Direct source for Shermer's and Hoffmann's open conjectures on orthogonal polygons with holes using vertex guards.

Schuchardt–Hecker [23].
NP-hardness source for art-gallery problems in orthogonal polygons.

Worman–Keil [30].
Decomposition and visibility-graph machinery for the orthogonal art-gallery problem.

Bjørling-Sachs [5].
Tight edge-guard theorem for rectilinear polygons; relevant when comparing 2D orthogonal edge guards with 3D edge-guard conjectures.

Katz–Roisman [16].
NP-hardness for guarding vertices of rectilinear domains by vertex, boundary, or point guards.

Polyomino, Perimeter, and Lattice Variants

Maßberg [17].
Primary peer-reviewed source for perimeter-over-six in hole-free polyominoes.

Maßberg habilitation [18].
Source for Conjecture 6.10 and the conditional hole extension. This is the key source for saying the holes case is not yet proven.

Biedl–Irfan–Iwerks–Kim–Mitchell [3].
Tight $\lfloor (m+1)/3 \rfloor$ area theorem for polyominoes, holes allowed, plus complexity context.

Diaz-Banez et al. [8].
Recent tight theorem for ortho-unit polygons and source of the integral orthogonal polygon $\lfloor N/6 \rfloor$ conjecture.

Pinciu [22].
Polyform extension including polyominoes, polycubes, and polyhypercubes. Relevant for discrete 3D analogues.

Filtser–Krohn–Nilsson–Rieck–Schmidt [10].
Current work on k-hop visibility in polyominoes, including open approximation questions for polyominoes with holes.

3D Orthogonal Polyhedra

Paterson–Yao [21].
Binary-space partition result used for the $\Theta(n^{3/2})$ point-guard bound in orthogonal polyhedra.

Viglietta thesis [27].
Central survey and source for guarding/searching polyhedra, the point-guard obstruction, and Urrutia-type edge-guard conjectures.

Benbernou et al. [2].
CCCG source for improved edge-guard bounds in orthogonal polyhedra: asymptotic $(11/72)m$ and $(7/12)r$.

Viglietta 2-reflex paper [29].
Proves optimal-style reflex-edge bounds for 2-reflex orthogonal polyhedra and gives an $O(n \log n)$ guard-placement algorithm.

Viglietta face-guarding paper [28].
Face-guarding results for several classes of polyhedra, including orthogonal subclasses.

Cano–Toth–Urrutia–Viglietta [6].
General edge-guard result in three-space and explicit open gap.

Algorithms and Related Visibility Models

Motwani–Raghunathan–Saran [19].
Perfect-graph approach to covering orthogonal polygons with star polygons.

Ghosh [12].
General approximation-algorithm context for art-gallery problems.

Durocher et al. [9].
Sliding-camera model for orthogonal art galleries, including holes.

Biedl et al. [4].
Sliding k-transmitter hardness and approximation results for orthogonal polygons.

11 Working Conclusions

1. The theorem "$\ell/6$ guards suffice" should be cited as proven only for *hole-free polyominoes*. The polyomino-with-holes version is conditional on Maßberg's rectangle-packing conjecture and should be described as open unless a new proof is supplied.

2. The broad integral statement "an integral orthogonal polygon of perimeter N is guardable by $\lfloor N/6 \rfloor$ guards" is a current conjecture, not a theorem.

3. The right 3D terminology is *orthogonal polyhedron* or *orthogonal polytope*. "Orthonormal" should not be used for the gallery object.

4. A 3D point-guard analogue of the planar perimeter theorem is blocked by the $\Theta(n^{3/2})$ point-guard bound. A serious 3D project should choose edge guards, reflex-edge guards, face guards, or a discrete polycube visibility model explicitly.

5. The strongest project targets suggested by this review are: prove or refute Maßberg's hole-packing conjecture; prove or refute the integral $\lfloor N/6 \rfloor$ conjecture; or make progress on the Urrutia $m/12 + O(1)$ and reflex-edge conjectures for general orthogonal polyhedra.

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